Intelligent System for Pakistan Sign Language Learning and Translation

Final Year Project - Progress Report

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Alizah Basit (22K-5155), Amna Mansoor (22K-5159), Hafsa Salman (22K-5161)

BS in SE

Department of SE

Fast School of Computing

Fast National University, Karachi Campus

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1. INTRODUCTION

The proposed project, **Pakistan Sign Language (PSL) Translator**, aims to bridge the communication gap between hearing-impaired individuals and the general public by leveraging modern Artificial Intelligence and Machine Learning techniques. The system is designed as a comprehensive platform that not only translates text and Pakistan Sign Language but also provides learning and assessment modules to promote inclusivity and accessibility.

The core idea of the project revolves around an encoder-decoder architecture, where the system takes signs and generates signs respectively. The system integrates multiple domains, including Deep Learning (DL), and Computer Vision, to ensure accurate and real-time translation.

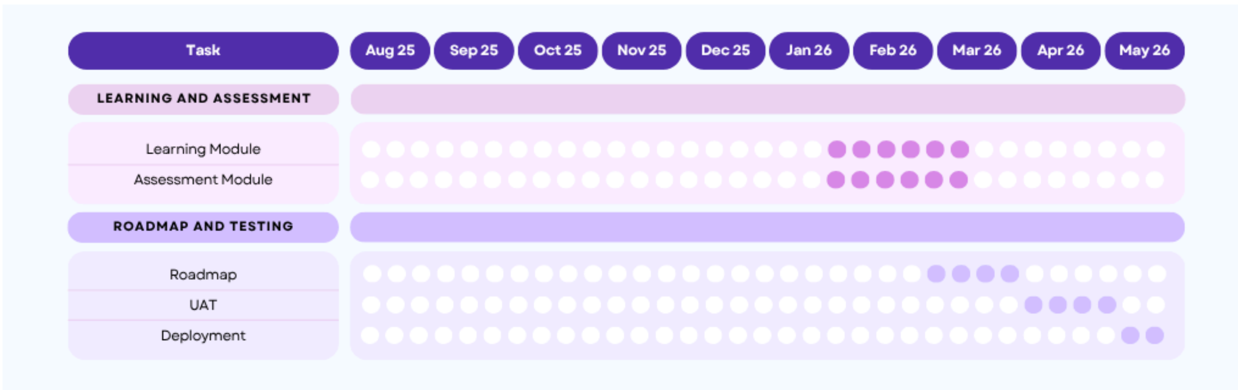
The platform consists of three main modules:

1. **Translation Module:** Converts spoken or written language into PSL and PSL gestures into text.
2. **Dictionary Module:** Development of a structured PSL dictionary, organized alphabetically and chapter-wise
3. **Learning Module:** Helps users learn PSL through structured lessons and interactive content.
4. **Assessment Module:** Evaluates user progress and provides feedback to enhance learning outcomes.

Additionally, the system includes features such as user authentication, progress tracking, and personalized learning paths. The dataset is prepared through data collection and augmentation techniques to improve model accuracy and robustness.

This project contributes to social good by promoting accessibility, inclusivity, and digital empowerment for the deaf and hard-of-hearing community in Pakistan. By combining advanced AI models with user-friendly interfaces, the system aims to deliver an efficient, scalable, and impactful solution.

2. TIMELINE



3. PROGRESS

3.1. MILESTONE 1 PROGRESS

During this phase, foundational work was completed, including:

- Literature review of existing sign language translation systems
- Software Requirements Specification (SRS)
- Software Design Specification (SDS)
- Initial UI/UX wireframes

This milestone established a strong theoretical and design base for the project.

3.2. MILESTONE 2 PROGRESS

In this phase, implementation and dataset-related tasks were completed:

- Data collection for PSL gestures
- Data preprocessing and augmentation
- Development of authentication system
- Initial implementation of dictionary and translation modules

This phase ensured that the system had sufficient data and basic functionality to proceed toward intelligent model integration.

3.3. MILESTONE 3 PROGRESS

Current progress includes:

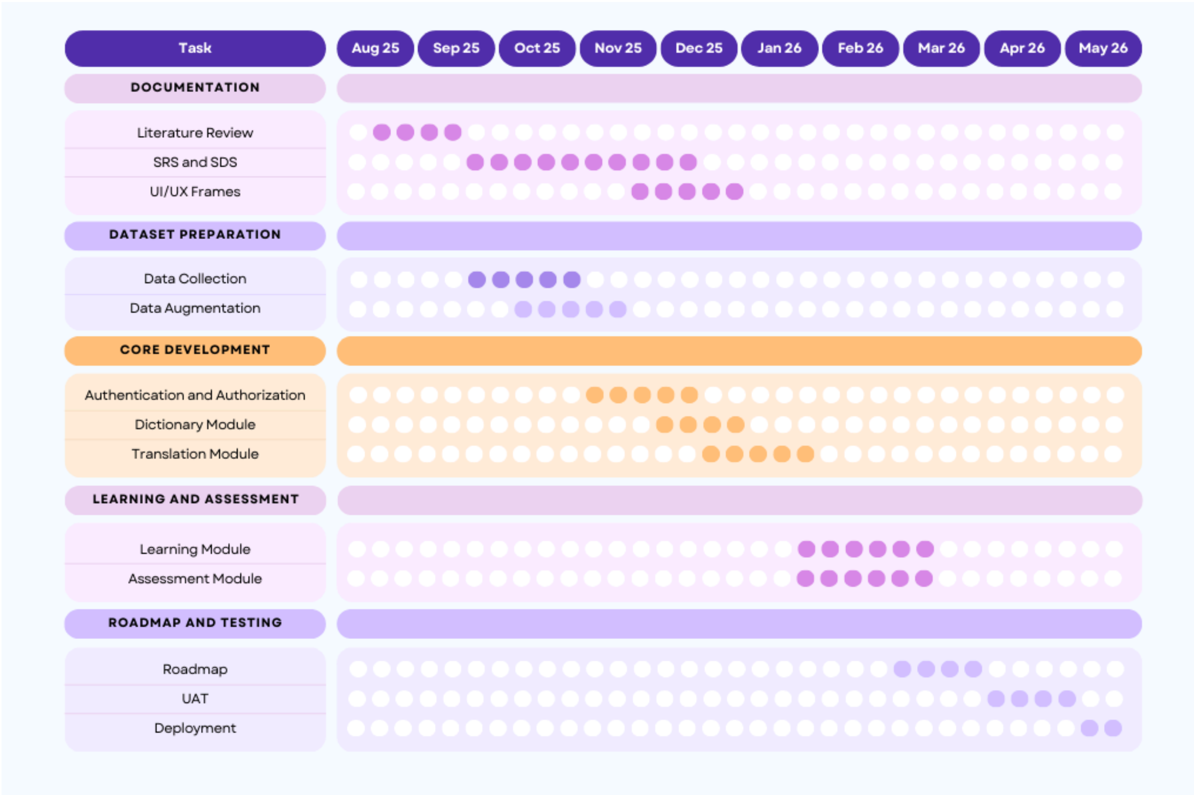
- Development of learning and assessment modules
- Integration of system components
- Training of multiple models for translation

The project is currently focused on training and evaluating three different deep learning models to improve translation accuracy and robustness:

1. **LSTM (Long Short-Term Memory)** – used for sequence modeling and capturing temporal dependencies in sign language data.
2. **Transformer-based Encoder Model (3 Encoder Layers)** – inspired by the research *"Sign Language Transformers: Joint End-to-end Sign Language Recognition and Translation"* by Necati Cihan Camgöz et al., the model employs self-attention mechanisms to capture temporal dependencies across video frames, enabling effective modeling of dynamic hand movements for isolated sign classification.
3. **Spatial-Temporal Graph Convolutional Network (ST-GCN)** – based on *"Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition"* by Sijie Yan et al., this model focuses on learning spatial and temporal features from skeletal movement data.

At present, the project is in the **model training and evaluation stage**, where performance improvements and accuracy enhancements are the primary focus before deployment.

4. UPDATED TIMELINE



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1. N. C. Camgöz, O. Koller, S. Hadfield, and R. Bowden, "Sign Language Transformers: Joint End-to-End Sign Language Recognition and Translation," in *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Seattle, WA, USA, 2020, pp. 10020-10030, doi: 10.1109/CVPR42600.2020.01004.
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